Server2 runtime services

S20b operator reference | source wiring only | 2026-10-17

Normal startup with ordinary **push** selection now starts supervised delivery after **both listeners bind**. Development verification uses the same production factory and stop path with owned loopback peers and synthetic keys. Do not run production startup with real credentials/endpoints during verification.

Select and serve

SERVER2_NATIVE_FAMILIES retains push and notice beside existing native families. One bounded S3 transport/cache is shared by registrations, claims/results, warning store/feed and notices. Workers dispatch through typed NativeFetch at the logical server2.internal origin, preserving units, full weather Accept-Language and v000902 paths. SERVICE_SERVER/API_SERVER and legacy HTTP forwarding do not select worker destinations. No constructor weather precollection, OAuth, FCM or paid Translator call occurs.

Alarm and conditional work begins at the next observed UTC minute; conditional minutes remain :07/:17/:35/:50. Stale queued ticks are rejected before claim work. No missed-minute catch-up occurs. Warning polls serialize every five seconds; their first successful baseline is silent. Acquisition remains request driven.

Registration and notices

Existing registration APIs remain POST /v000705/push, PUT/DELETE /v000902/push and POST /v000902/push-list. Success follows durable journal publication. Quiesce rejects new mutations with existing storage-error responses; queued writes may settle or remain incomplete. GET/HEAD /v000903/notice/latest retains language selection, public max-age=300 and Vary: Accept-Language. Missing publication is notice:null; unreadable/corrupt cold storage is 503/no-store. Reads never translate.

Final send boundary and recovery

Claims are durable before FCM POST and are never released for automatic resend after failure, ambiguity or cancellation. After OAuth, alarm checks the current subscription/minute, conditional checks committed alert state/time, and warning checks subscription/time/current zones. Stop is checked again after awaited guards, immediately before POST. Stop-before-POST is **NotSent**; a lost/timed-out POST response is **Unknown**. Timestamp-only refresh still permits delivery. An edit after POST admission remains protected through stale cleanup; the push may already have happened.

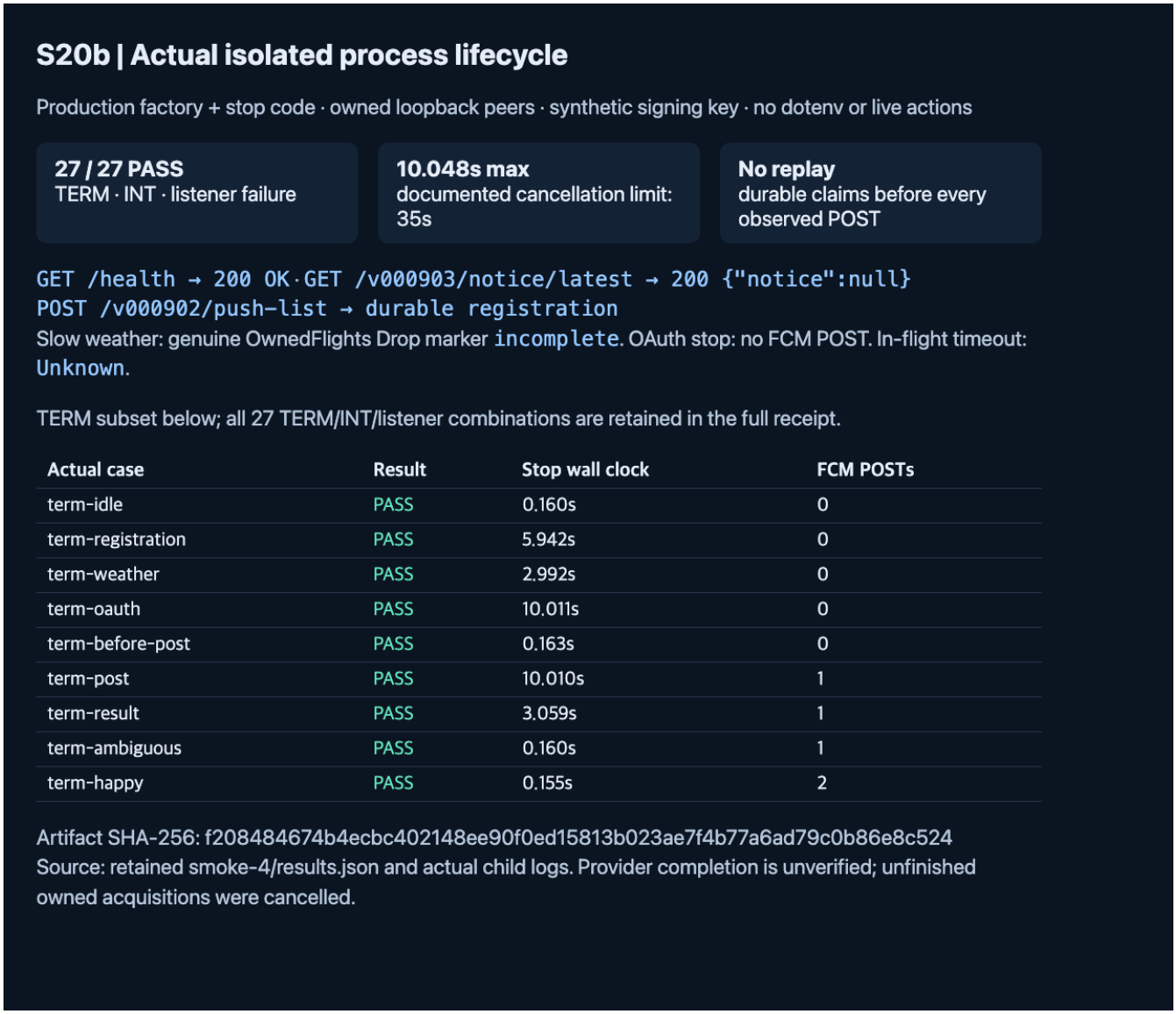
Stop: one cancellation envelope

SIGTERM, SIGINT and listener failure share the selected stop path: with a push writer or delivery supervisor, 34 seconds for concurrent HTTP/job/writer drain plus at most one second runtime teardown (35-second maximum). With neither present, HTTP keeps its foundation 5-second drain plus at most one second teardown. A signal deadline logs the original forced-stop message; listener failure remains an error. Weather waiters are 3 seconds each, OAuth/FCM 10 seconds each, ledger operations 4 seconds, writer S3 operations 2 seconds, and a writer guard may queue up to 9 seconds. Multiple operations or large batches can exceed the envelope; cancellation retains claims and marks work incomplete. Joined handles do not mean successful delivery or complete provider acquisition. Owned native acquisitions can outlive their waiter; unfinished async work is cancelled at teardown and partial publication must be recovered from S3.

Credential files remain lazy per product. Missing/invalid files fail only that product's send, not startup; retired TodayAir is no new mandatory account. Future deployment readiness checks actual required files separately. Deployment, live sending, EC2/quota and seven-day shadow remain excluded.

Actual isolated verification

Production factory + OS signals + owned HTTP peers



The browser capture above comes from retained actual smoke results. All 27 TERM/INT/listener-failure combinations passed; maximum observed stop was **10.048 seconds**. It includes idle, queued registration, weather/OAuth waits, final pre-POST guard, in-flight POST, result write, ambiguity and scheduled success. Genuine OwnedFlights task Drop emitted an incomplete-cancellation marker. In-flight timeout remained Unknown. Restart did not replay claims. Additional actual checks proved second-bind failure starts no delivery and two overlapping processes produce one POST.

Historical evidence: delivery **60 PASS**, runtime **21 PASS** plus one ignored child entrypoint; retained state **18 PASS** and notices **24 PASS**. all-target Clippy and placement passed. CI and eligible independent review remain separate exact-head gates. Latest source 411eb9a8 restored HTTP-only forced stop; held push OAuth stop was 10.036 seconds with zero POST. Preliminary 2-second-peer results, intended Red (stale POST=1), Green (POST=0), setup/oracle failures and corrected rounds are retained; none was relabeled.

Reproduce with the configured toolchain/shared lease, retain the runtime_services test executable and hash inside the lease, then run tools/runtime_services_smoke.py with --binary and --out. The child environment is explicitly cleared and loads no dotenv. Source/manual: docs/operations/server2-runtime-services.md. Selected evidence: docs/evidence/tasks/issue-2705-s20b/verification.md.